

Topic	Subtopic
Review of Random Variables	Random variable definition, Discrete vs Continuous, Expected value $E[X]$, Variance $\text{Var}(X)$
Uniform Continuous Random Variable	PDF curve, CDF curve, Expectation, Variance, Properties, Sample problems
Exponential Random Variable	PDF curve, CDF curve, Expectation, Variance, Memoryless property, Relation with Poisson, Relation with Erlang and Gamma distributions, Sample problems
Poisson Distribution	PMF, Expectation, Variance, Event-count interpretation, Relation with Exponential distribution, Sample problems
Geometric & Pascal / Negative Binomial	PMF, Expectation, Variance, Relation with Exponential and Poisson, Sample problems
Erlang Distribution	PDF, CDF, Relation to Exponential, Expectation, Variance, Sample problems
Gamma Distribution	PDF, CDF, Expectation, Variance, Gamma function, Relation with Erlang, Sample problems
Beta Distribution	PDF, CDF, Properties, Sample problems
Gaussian / Normal Distribution	PDF curve, Standard Normal, Z-table for probabilities, Standardization, Sample calculations
Bivariate / Multivariate Random Variables	Pair of random variables, Joint random variables, Joint CDF (JCDF), Joint PDF (JPDF), Marginal probability, Covariance, Correlation, Multivariate random variables, Multinomial random variable, Sample problems